

Patient Safety and Technology

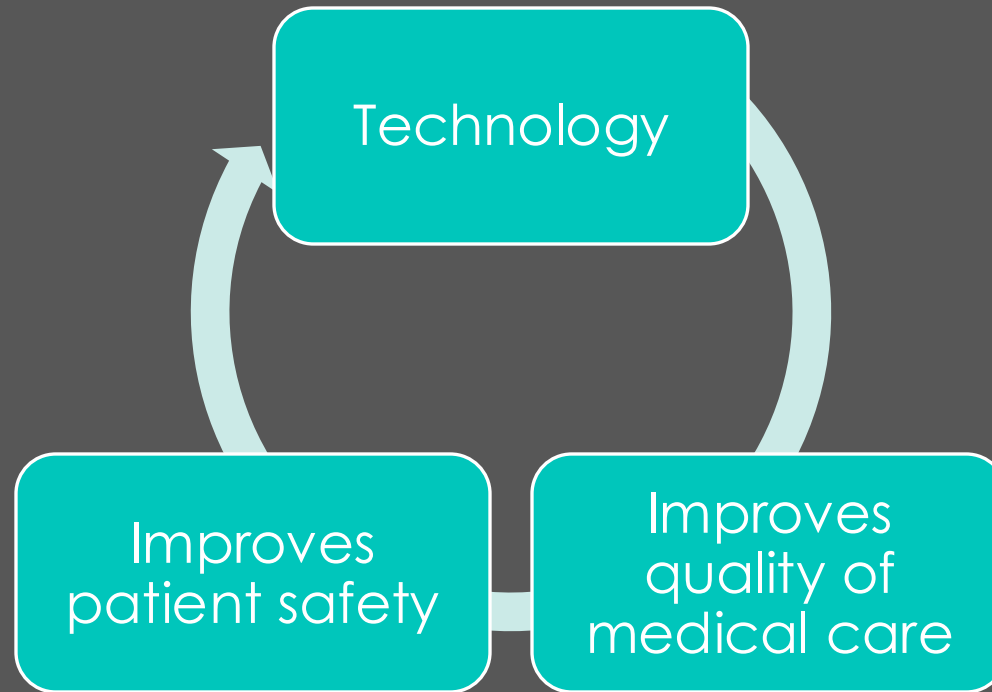
DR. SHANTHI VANKA

Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals,
3rd Edition, 2009

Lecture learning outcomes:

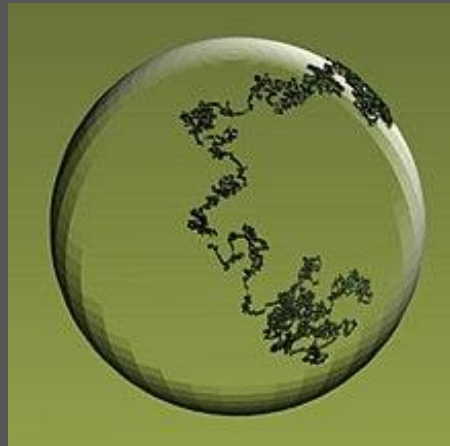
- List the various technologies that have a potential to decrease Medication Errors
- Explain the application of information technology in improving patient safety

Basic premises of this topic is:



Era of Brownian motion in health care

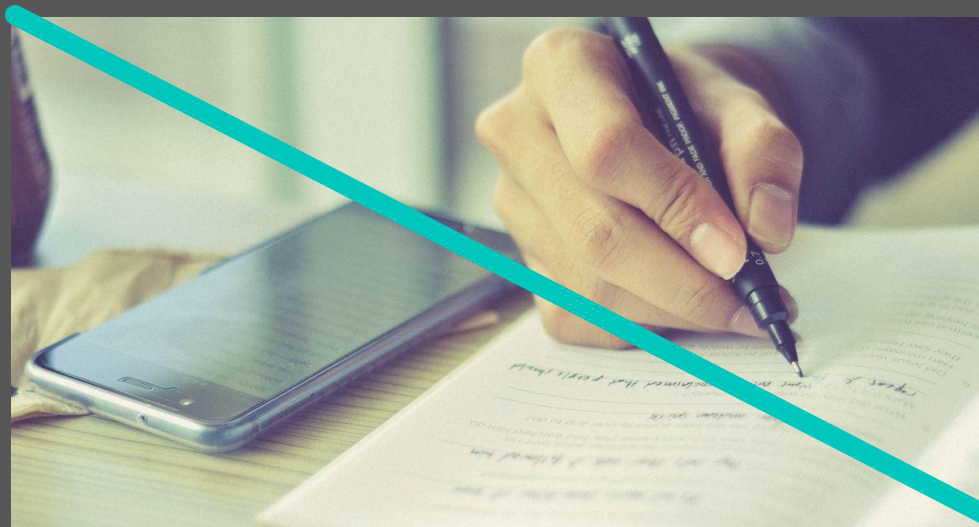
- **Institute of Medicine (IOM)** report To Error is Human estimated that many **inpatients die** every year due to **preventable errors**.
- The **current medical system** was described as an “**era of Brownian motion in health care**”.



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 232.

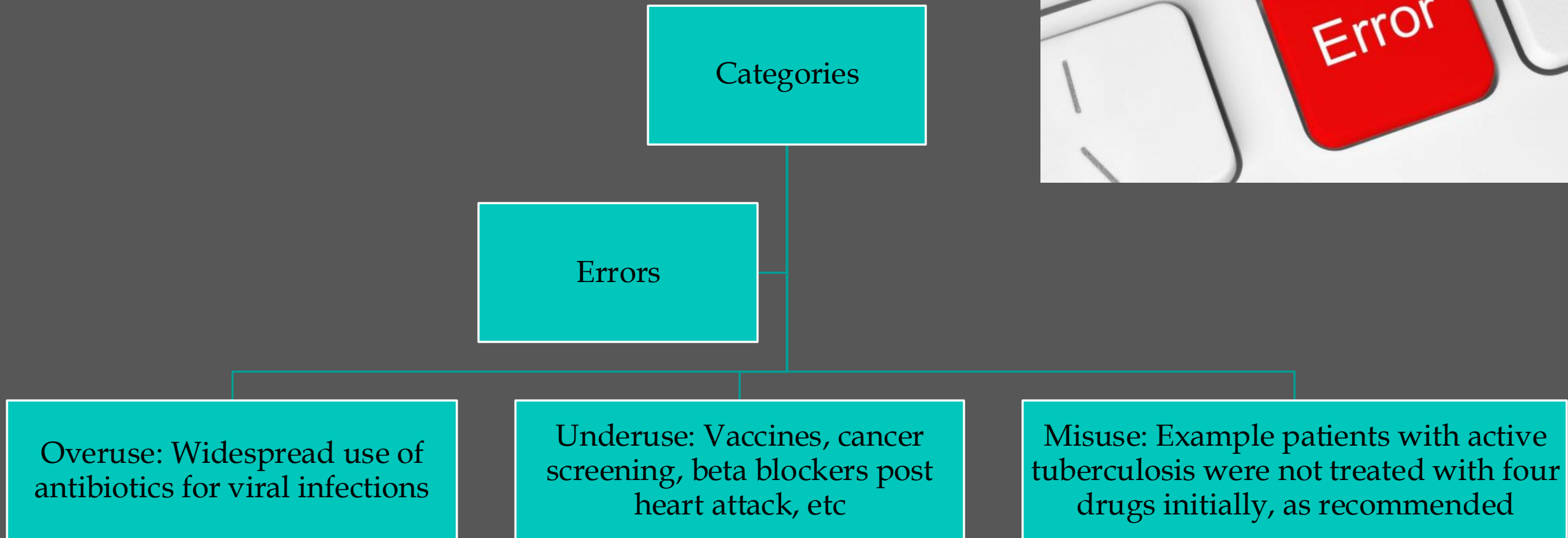
IOM Executive Summary recommended

- Create a **national information infrastructure** to improve health care delivery and research.
- Goal is to **eliminate handwritten notes.**

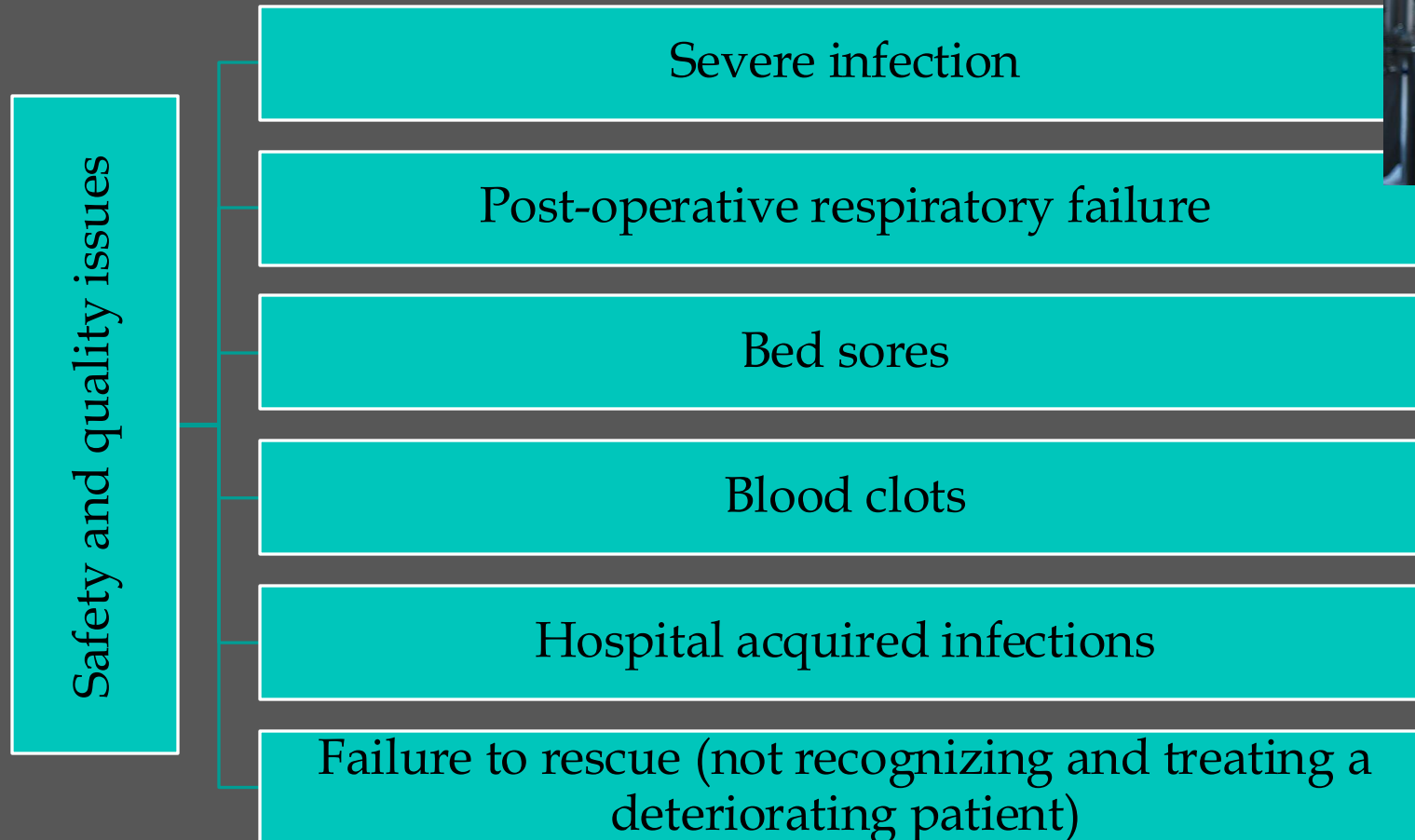


- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 232.

What are the reasons for different categories of medical errors?



The leading 6 safety and quality issues measured were:



Barriers to Improving Patient Safety through Technology

1. Organizational

2. Financial

3. Trained HIT worker

4. Privacy

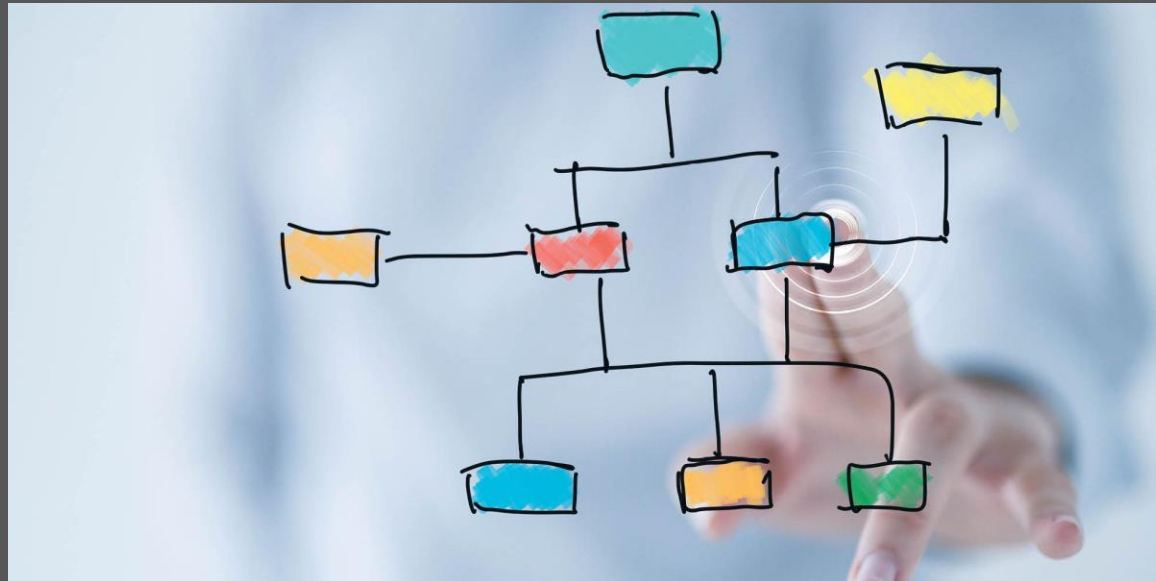
5. Security

6. Behavioral

7. Error reporting

1. Organizational

- Many small physician groups have **no loyalty to hospitals** or other healthcare organizations. They do not interact with other physician offices or healthcare organizations and **do not share data**.



2. Financial

- Who will **pay** for what?



3. Trained HIT Workforce

- How can we train and implement technologies such as electronic health records for everyone without **expanding the number of health IT workers?**



4. Privacy

- It is a state in which **one is not observed** or **disturbed** by other people.



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 235.

5. Security

- Several major episodes of **identity theft** have alarmed patients and staff and have contributed to the **slow adoption of electronic health records** and other technologies.



6. Behavioral

- **Lack of the use of Computerized physician order entry (CPOE).**



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 235.

7. Error reporting

- **Reporting** continues to be **voluntary** and **inadequate** at best.



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 235.

Patient Safety and Quality Improvement Act

- This act is limited by the fact that there is **no definition of “error”** and **data** can only be used in a **criminal case** and with the **physician’s permission**.



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 236.

Health goals that have IT implications



1. Improve the **accuracy** of **patient identification**
2. Improve the **effectiveness** of **communication** among caregivers
3. Improve the safety of **using medications**
4. Accurately and completely **reconcile medications** across the continuum of care

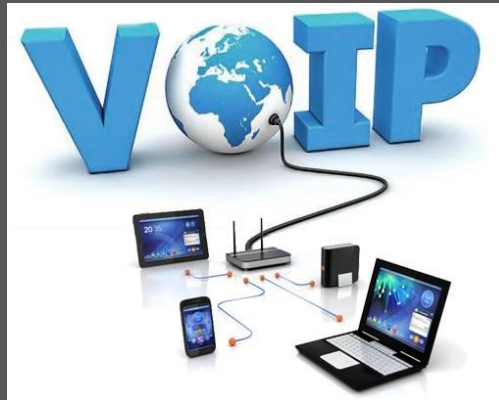
1. Improve the accuracy of patient identification:

- Potential for **bar coding**



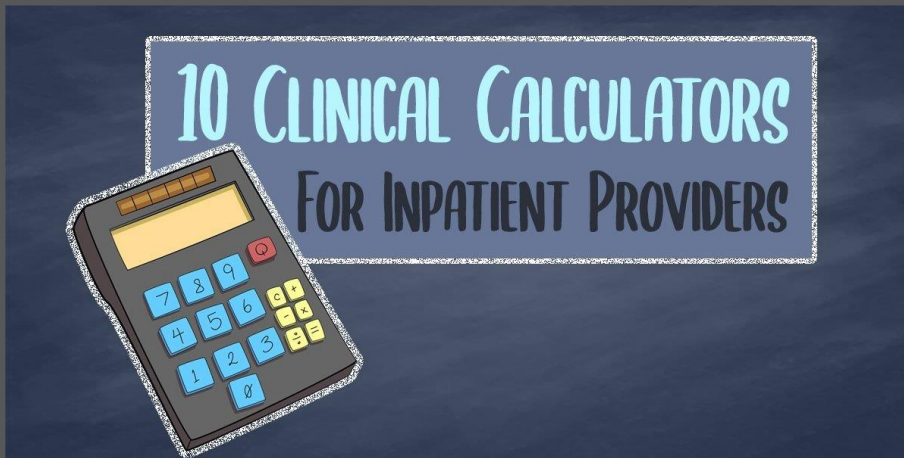
2. Improve the effectiveness of communication among caregivers:

- **Expedite critical test results** to clinicians by using **cell phones or voice over Internet protocol (VoIP)**.
- Other examples are **patient portals, telemedicine/teledentistry** and health information organizations



3. Improve the safety of using medications:

- Examples could be the use of **online or smartphone drug programs** as well as clinical calculators



4. Accurately and completely reconcile medications

- If all pharmacy **records** were **online** and **linked by a health information organization** it would be easy to reconcile all medications



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 236.

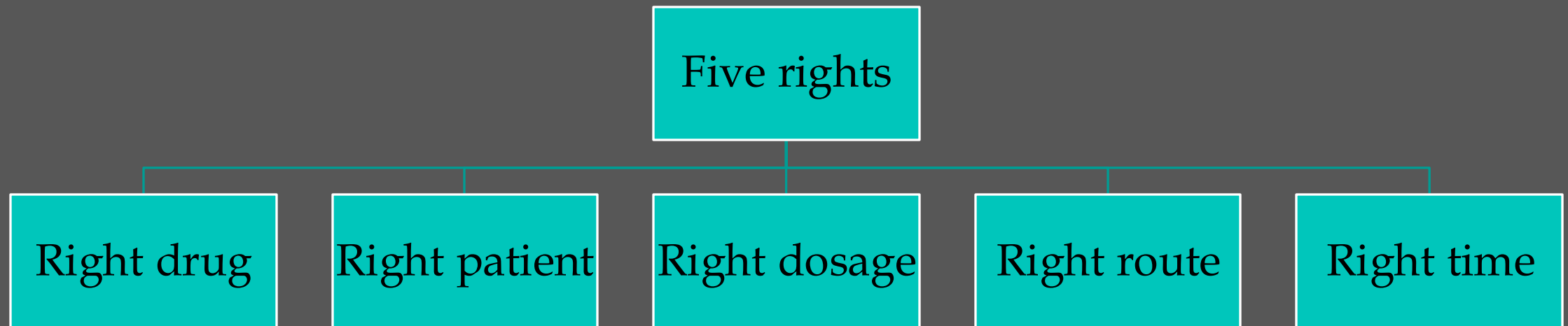
Information Technology and Patient Safety

- **Medication error reduction** remains the most important patient safety area impacted by healthcare IT.



“Five rights” before medication administration:

- Safety organizations mandate that patients receive the “five rights” before medication administration
- Fortunately, **99% of medication errors** do **not** result in **injury**.



Technologies That Have the Potential to Decrease Medication Errors:

Technologies to decrease errors

Computerized Physician Order Entry

Automated Inpatient Medication
Dispensing Devices

Home electronic medication management
systems

Pharmacy Dispensing Robots

Electronic Medication Administration
Record (eMAR)

Technologies to decrease errors

“Smart” Intravenous (IV) Infusion Pumps

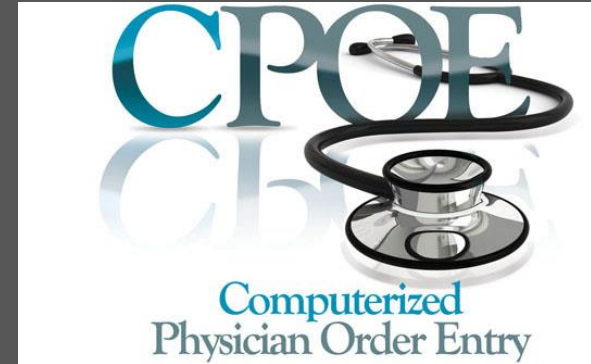
Calculators

Bar coding and RFID

Medication Reconciliation

Electronic Prescribing

Computerized Physician Order Entry (CPOE)



Inpatient CPOE

Outpatient CPOE

Clinical Decision Support

Accurate Drug Histories

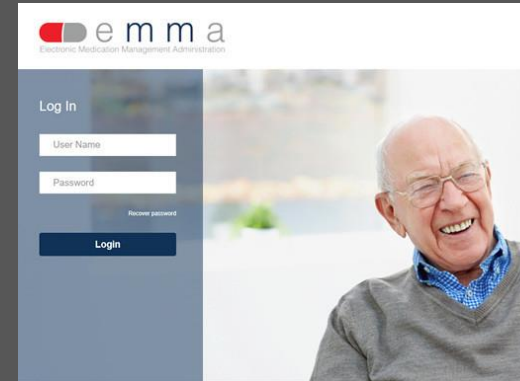
Automated Inpatient Medication Dispensing Devices



- Like ATM machines, these devices communicate with pharmacy computers and dispense medications stocked by the pharmacy
- Device keeps medication records

Home electronic medication management systems (EMMA)

- Mainly for the elderly at home
- Medications are loaded into the machine
- The device is connected to the pharmacy via the Internet so they can monitor compliance and adjust doses
- The device (EMMA) gives a visual and audible alert when it is time to take a medication



Pharmacy dispensing robots

- Robots are very helpful when there is a shortage of pharmacists or staff. Pharmacists to have more of a supervisory role
- Ideally, systems would receive electronic prescriptions from outpatient and inpatient areas, then be checked by both the EHR (Electronic Health Records) and the pharmacist, then labels are printed and the prescription filled



Electronic Medication Administration Record (eMAR)

- No need to search for the patient's chart to see what medications the patient is on
- Application is available to nurses and others, like physicians making rounds



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 241.

“Smart” Intravenous (IV) Infusion Pumps

- Smart pumps can be programmed to deliver the correct amount of IV drugs and are associated with drug libraries and alerts that the dose differs from hospital guidelines. This feature is known as a “dose error reduction system” (DERS).



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 242.

Calculators

- Johns Hopkins University created a web based paediatric total parenteral (IV) nutrition (TPN) calculator and as a result reduced medication errors



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 242.

Bar coding and RFID

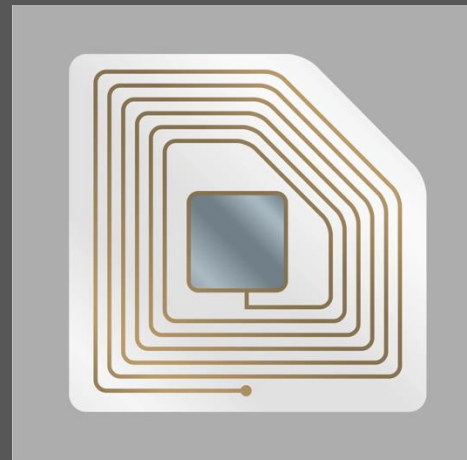
- Bar coded medication administration (BCMA) involves a variety of elements: bar code printers, scanners, a network (wired or wireless) to connect to a server, server with bar coding software and integration with the pharmacy information system.



- Chapter 14. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 244.

Bar coding and RFID

- Radio Frequency Identification (RFID): tags can be read if wet or thru clothing. Better for blood and IV bags
- Can track patients within a hospital with an active tag that works like a transmitter and gives location and time



RFID Chip

- The chip contains a 16 digit number that is read by a scanner. The number is used to locate the web based medical record.



Medication Reconciliation

- When patients move from hospital-to hospital, from physician-to-physician or from floor-to-floor, medication errors are more likely to occur.
- If all medical offices, pharmacies and hospitals were connected to a shared health information organization, then the answer would be simpler and electronic.



Electronic prescribing

- E-prescribing is available in two modes: 1) a standalone software program installed on a smartphone, computer or available as a web-based application 2) eRx integrated into an electronic health record.



- Chapter 15. Robert E. Hoyt. Medical informatics-Practical guide for health care professionals, 3rd Edition, 2009. Page 256.

Conclusion

- IT adoption and patient outcome measures and concluded there was a definite correlation.



Thank you